

11. Performance Analysis

11.1 Introduction

Performance analysis involves evaluating the results obtained from load test execution to determine how efficiently the application performs under simulated user load conditions.

The BlazeDemo Flight Booking Application was tested using Gatling with 20 virtual users ramped up over 30 seconds. Three different business scenarios were executed to analyze system behavior, response times, throughput, request success rates, and overall application stability.

The performance metrics collected during execution provide valuable insights into the application's ability to process user requests and maintain acceptable performance levels under load.

11.2 Response Time Analysis

Response time is one of the most important performance metrics in load testing. It represents the time taken by the application to receive a request, process it, and return a response to the user.

Lower response times indicate better application performance and user experience.

Scenario 2 – Choose Flight

Metric	Value
Minimum Response Time	357 ms
Maximum Response Time	756 ms
Average Response Time	476 ms
95th Percentile	726 ms
99th Percentile	742 ms

Analysis

The average response time of 476 milliseconds indicates that the application responded quickly during flight selection operations.

The maximum response time remained below one second, which is considered acceptable for web applications.

All requests completed within 800 milliseconds, demonstrating consistent performance throughout execution.

Scenario 3 – Complete Booking

Metric	Value
Minimum Response Time	353 ms
Maximum Response Time	1620 ms
Average Response Time	516 ms
95th Percentile	1163 ms
99th Percentile	1592 ms

Analysis

The Complete Booking scenario recorded higher response times because it involved multiple business transactions, including:

- Flight Search
- Flight Selection
- Passenger Information Submission
- Booking Confirmation

The average response time remained close to 500 milliseconds, indicating good overall performance.

Although a small percentage of requests exceeded one second, the application continued to process all transactions successfully.

Observation

Response times increased as transaction complexity increased.

Scenario 3 required additional server-side processing compared to Scenario 2, resulting in slightly higher response times.

Overall Response Time Evaluation

The application maintained acceptable response times across all tested scenarios.

Key observations:

- Average response times remained below one second.
- Most requests completed within acceptable limits.
- No severe performance degradation was observed.
- Response times increased proportionally with workflow complexity.

The response time analysis indicates that the BlazeDemo application is responsive and capable of handling the configured workload efficiently.

11.3 Throughput Analysis

Throughput measures the number of requests processed by the application within a given time period.

Higher throughput indicates better request processing capability.

Scenario 2 Throughput

Metric	Value
Requests per Second	1.67 req/sec

Analysis

The application processed approximately 1.67 requests per second during the flight selection workflow.

The throughput remained stable throughout execution and no request backlog was observed.

Scenario 3 Throughput

Metric	Value
Requests per Second	2.05 req/sec

Analysis

The Complete Booking scenario achieved a throughput of approximately 2.05 requests per second.

Even though this scenario involved more business transactions, the application successfully processed all incoming requests.

Overall Throughput Evaluation

The throughput results indicate that the application effectively handled incoming traffic without request congestion.

Observations:

- Stable request processing was observed.
- No transaction delays caused by request accumulation.

- The server processed all requests successfully.

The throughput values demonstrate that the application can sustain the configured user load while maintaining stable performance.

11.4 Success Rate Analysis

Success rate represents the percentage of requests completed successfully during test execution.

A high success rate indicates application reliability and stability.

Scenario Results

Scenario	Success Rate
Search Flights	100%
Choose Flight	100%
Complete Booking	100%

Analysis

All requests executed successfully across all scenarios.

No transaction failures occurred during test execution.

The application successfully processed:

- Flight Search Requests
- Flight Selection Requests
- Booking Transactions

without generating any errors.

Observation

A 100% success rate indicates excellent application reliability under the configured load conditions.

The results confirm that the BlazeDemo application remained stable throughout testing.

11.5 Error Analysis

Error analysis evaluates failed requests, exceptions, server errors, and transaction failures observed during execution.

Results

Metric	Value
Failed Requests	0
HTTP Errors	0
Application Errors	0
Transaction Failures	0

Analysis

No errors were encountered during any test execution.

All requests returned successful responses and passed validation checks.

The following validations were successful:

- Home Page Access
- Flight Search
- Flight Selection
- Booking Submission
- Confirmation Page Generation

Observation

The absence of errors indicates that the BlazeDemo application handled all simulated user requests successfully.

No instability, crashes, or unexpected behavior were observed during testing.

11.6 User Load Handling Capacity

User load handling capacity refers to the application's ability to process concurrent user requests while maintaining acceptable performance levels.

During testing:

- 20 Virtual Users were simulated.
- Users were introduced gradually over 30 seconds.
- Multiple business transactions were executed simultaneously.

Application Behavior Under Load

The application successfully:

- Accepted all incoming requests.
- Processed concurrent transactions.
- Maintained stable response times.
- Returned successful responses.
- Completed all booking workflows.

Observations

The BlazeDemo application demonstrated the ability to support the configured workload without performance degradation.

Even during the Complete Booking scenario, which generated the highest workload, the application remained stable and completed all transactions successfully.

Capacity Assessment

Based on the executed tests:

- The application comfortably handled 20 concurrent virtual users.
- No request failures occurred.
- No resource-related issues were observed.
- System responsiveness remained within acceptable limits.

The test results indicate that the BlazeDemo application can efficiently manage the configured user load while maintaining performance and reliability.

11.7 Overall Performance Assessment

Based on the collected performance metrics, the BlazeDemo Flight Booking Application demonstrated stable and reliable performance throughout all load testing scenarios.

Key findings include:

- 100% request success rate.
- Zero transaction failures.
- Acceptable average response times.
- Stable throughput values.
- No application errors.
- Successful handling of concurrent user activity.

The performance analysis confirms that the application successfully handled the configured workload and maintained consistent performance across all tested business workflows.